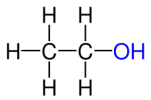
SEE 2081 (2025)

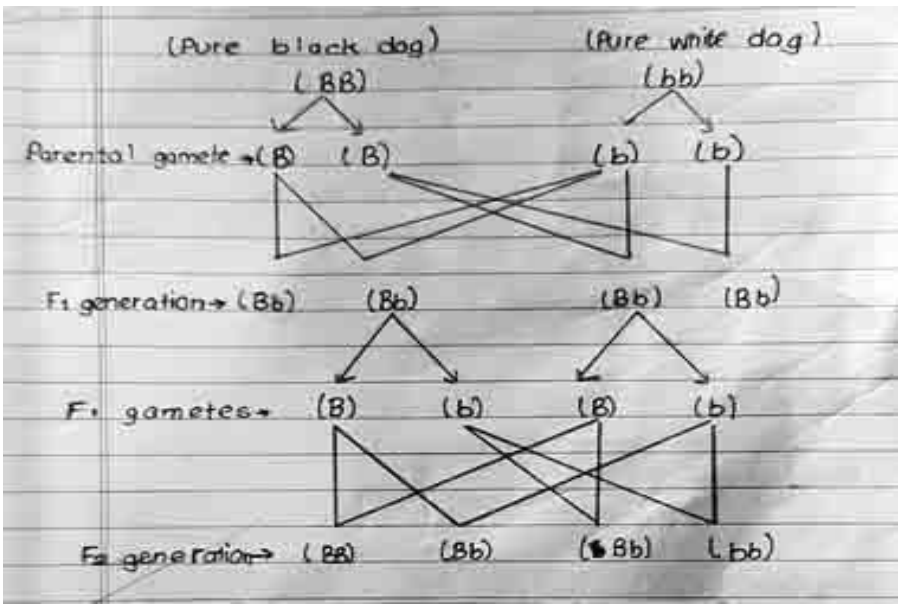
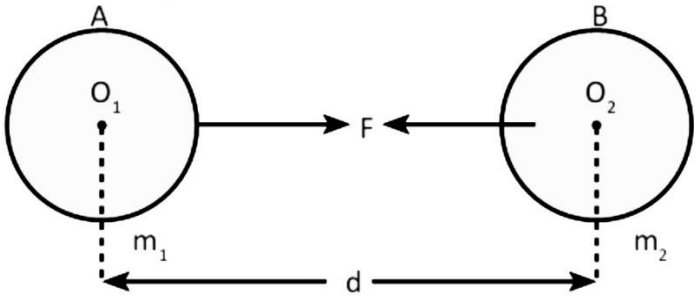
अनिवार्य विज्ञान तथा प्रविधि

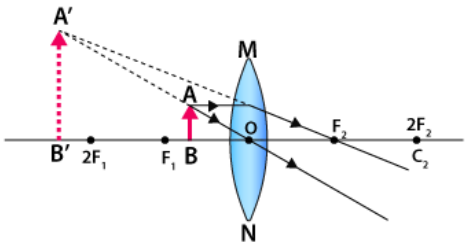
उत्तरकुन्जिका

o; s'lGhsfdf ePsf] afx]s cGo ;xL j}slNks tl/sfaf6 ;d:of ;dfwfg u/]df klg c+s k|bfg ug'{kg]{ 5 . pQ/s'l~hsfdf k|To]s r/0fsf] k|fKtf+Í ! k|bfg ug'{kg]{5 tyf PsfO 5'6 ePdf)=% cª\s dfq 36fpg'kg]{ 5 .

k|= g+=	pQ/	cª\s
	;d"x 's'	
1) a.	(iii) Newton	1
1) b.	(ii) Due to locomotion with the help of tube feet	1
1) c.	(iii) Leukopenia	1
1) d.	(i) symbiotic	1
1) e.	(ii) 30N	1
1) f.	(ii) Convex lens	1
1) g.	(iii) When average density and critical density are equal	1
1) h.	(iii) $NaNO_3 + H_2O$	1
1) i.	(iv) Mixture of carbondioxide and oxygen	1
1) j.	(iii) A and D	1
	;d"x ...v'	
2) a.	the reputation of a person or organization in digital platform(or other correct answer)	1
2) b.	Shark	1
2) c.	Pulmonary vein	1
2) d.	The influenced area of magnet in which magnet attracts the magnetic materials (or other correct answer)	1
2) e.	Because it tells about the origin by the explosion of the supernova/atom which is more reliable than other theories.	1
2) f.	$2Na + Cl_2 \rightarrow 2NaCl$ (or any other correct answer)	1
2) g.	NaOH	1
2) h.	H_2CO_3	1
2) i.	Saponin	1
	;d"x ...u'	
3.	• The unit which is independent of other units is called fundamental units. • Example kg, m etc.	1 1
4.	• Because they have distinct root, steam and leaves • they have developed vascular tissues (or any other correct answer)	1 1
5.	**Drone bee** / **Worker bee** 1. Bigger than work bee / 1. Smaller the drone (smallest) 2. Body hairy, black / 2. legs hairy, mouthparts is chewing and lapping types 3. lazy / 3. laborious 4. They help to mate with queen been. / 4. They bring nectar, form hive, take care of larva's etc. **(Any two correct difference)**	1 1

6.	• The plants grow and form new plants quickly. • They have lots of different traits that are easy to see. (an related answer)	1 1
7.	(i) Meiosis (ii) Process P - occurs in reproductive cells, helps in reproduction Q - occurs in somatic cells, helps in healing of wounds (any one correct difference gets - 1 mark)	1 1
8.	– Aloe-vera – It is used to lighten the hair, for skin diseases to supply vitamin A.	1 1
9.	(i) 2N +1N = 3N (ii) Archimedes' principle	1 1
10.	(i) Water freezes, volume increases of solid state of water(ice) at 0^oc resulting expansion. (ii) pressure increases inside and brust wall of pipes.	1 1
11.	m = 500 gm = 0.5 kg, dt = 126° C, Q = 5980 J $S = \frac{Q}{m \times dt}$ $= \frac{5980}{0.5\times126} = 94.92$ J/kg^{o}C	1 1
12.	• The light that reaches eyes from different raindrops always exist at nearly the same angle. So, same thickness. • Horizontal condition of the earth (the lower half of the circle is obstructed by the horizon) or (correct answer like)	1 1
13.	• $Mg + CO_2 \rightarrow MgO + C$ • $2Mg + CO_2 \rightarrow 2MgO + C$	1 1
14.	**Mineral** / **Ore** (i) The compound of metal found in the crust of the earth are called mineral / (i) The mineral from which metal can be extracted. (ii) All minerals are not ores / (ii) All ores are mineral (iii)It consists metals or non-metals / (iii) Metal can be purified from this by metallurgy Any two correct differences gets 2 marks	1 1
15.	ethyl alcohol (ethanol) 	1 1
16.	• Non-biodegradable. • it causes chemical pollution (any related answer)	1 1
	;d"x …3'	
17.	**Positive Aspects** • Digital library • news paper • digital payments • online business • social networks (any two correct answer) **Negative aspects** • social behaviour (bad) • cyber crime	1 1 1 1

	• reduced physical material • lack of exercise (any two correct answer)	
18.	 Phenotype ratio = 3:1 genotype ratio = 1:2:1	
19.	(i) C-platelets counts is low (ii) A-RBC counts is low (iii) B-WBC count is high (iv) D-amount of sugar is high	1
20.	Force of attraction between two heavenly bodies is (i) directly proportions to the product of their masses and (ii) inversely proportional to the square of distance between their center  $F \propto m_1 m_2$ ------- (i) $F \propto \frac{1}{d^2}$ ---------- (ii) Combining the equation (i) and (ii) we get , $F \propto \frac{m_1 m_2}{d^2}$ or, $F = G \frac{m_1 m_2}{d^2}$ (Here G is a constant known is gravitational constant its value is 6.67×10^{-11} Nm^2/kg^2)	1 1 1 1 1
21.	A = Convex lens B = Concave lens	1 1

	A forms virtual and magnified image.  	2
22.	(i) X - step up Y - step down (ii) — electrical appliances cannot work properly. — electrical appliances may damage	1 1 1 1
23.	(i) physical and chemical properties of the elements are periodic function according of their atomic number (ii) Group VIA/ 16 a) $Mg + O_2 \rightarrow MgO$ b) $2Mg + O_2 \rightarrow 2MgO$	1 1 1 1

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